Identity Graph Logic (IGL) vs. Knowledge Graph Products

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Knowledge graphs are sold today as **data products** — Neo4j, Stardog, Amazon Neptune, Microsoft Graph and their peers organize entities and relationships so humans and models can *query what is known*. Identity Graph Logic is a **control plane**: the graph exists so that every AI action can be bound to *who is acting, under whose authority, within what coverage* — deterministically, at runtime.

Dimension	Knowledge Graph (as sold today)	Identity Graph Logic (IGL)
Core question	"What is known about this entity, and how is it connected?"	"Who is acting — **who am I, who are we** — under whose authority, and within what coverage?"
Primary artifact	Queryable graph of facts and entities (property graph / RDF), optimized for traversal and analytics	Governed digital representation of a person or company: identity, ownership, authority, delegation, resources, policies
Primary consumer	Analysts, search, recommendations, entity resolution, RAG context fed into LLMs	Runtime enforcement for AI models, agents, and MCP tools — the graph is consulted before every material action
Truth model	Asserted plus inferred edges; probabilistic enrichment and fuzzy matching are acceptable	Deterministic: authority is computed by UDM — explainable, reproducible, no inference at decision time
Output of a query	Records, paths, similarity scores	An allow/deny decision, bounded coverage, and an audit record tied to identity lineage
Identity semantics	People and organizations appear as *data subjects* — things described	People and organizations are *principals* — actors holding keys, delegations, and obligations
Change and revocation	Data updates propagate on ingestion cycles; eventual consistency is normal	Revocation propagates across models, agents, tools, and live sessions; keys reference current authority state
Delegation	Not native — delegation can only be stored as more data	First-class: user→agent and cross-graph delegation without merging graphs; derived authority can only be equal to or narrower than its source
Commercial buyer	Data engineering, analytics, and AI-context teams	Compliance / risk owner with the CISO — deterministic auditability is a compliance purchase
Relationship between them	A knowledge graph can serve as substrate or enrichment source for the Identity Graph	IGL is the enforcement layer a KG cannot become: authority computation, key semantics, and the AI+MCP enforcement chain

Bottom line. Knowledge graphs organize **what is known**. Identity Graph Logic governs **who may act**. A KG can inform the Identity Graph — it cannot replace deterministic authority computation, live key semantics, or runtime enforcement.

IOS+ (MCP Bridge / Gateway Server) vs. OpenRouter-Class AI Gateways

OpenRouter and similar AI gateways solved **model access**: one API, one credential, many providers, with routing, fallback, budgets, and logging. IOS+ is the **MCP Bridge / Gateway Server** in the IGL stack: it governs **action** — every tool call, data touch, and delegated operation — under authority computed from the Identity Graph. The two control points answer different questions.

Dimension	OpenRouter-class AI gateway	IOS+ — MCP Bridge / Gateway Server
Control point	Model requests: which LLM API gets called, at what cost, with which fallback	Action execution: which agent may invoke which tool, on which resource, under whose identity — plus which models may participate
What the key means	A credential carrying credits, quota, and rate limits	A scoped bearer key that **references live, graph-derived authority state** — not a static secret
Questions asked at runtime	Is this key valid? Can it call this model? Is it under budget?	Which identity does this key represent? Which graph governs it? What authority was computed? Which model version is authorized? May this model use this MCP tool? Is the delegation still valid — and from which graph did it originate?
Model governance	Catalog access across providers, routing, fallback, cost caps	Exact model-version activation bound to the identity's coverage; model participation is itself an authorization decision
Tool governance	Out of scope — AI gateways do not enforce MCP tool calls	Native: per-tool, per-parameter authority derived from the Identity Graph through the MCP bridge
Enforcement basis	Gateway policy configuration (limits, guardrails, DLP rules)	Deterministic authority computed by UDM **outside model reasoning** — the same boundaries on every approved model
Audit	Request/response logs for billing and observability	Deterministic, replayable action audit tied to identity lineage across models, tools, and sessions
Delegation	None — API keys are shared secrets, transferable by copy	User→agent and cross-graph delegation, attenuation-only, revocable, with propagation across sessions
Commercial buyer	Developers and platform engineering optimizing cost and access	Compliance / risk and platform teams operating governed AI — a compliance purchase, not an access purchase
Relationship between them	Can serve as an upstream model route underneath IOS+	IOS+ governs what a model gateway cannot see: tools, data, delegation, and the identity behind every call

Bottom line. AI gateways commoditize **access to models**. IOS+ governs **action under identity**. The categories will converge — gateways are already adding identity features — but identity bolted onto a gateway is policy configuration; identity at the foundation is architecture.